

MOD 308 Course Outline

MOD 308 — Multi-Regional Input-Output Analysis

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	35.004 (indicative band 30–40)
Shape	4 parts · 12 sessions · 8 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 302, for the Leontief inverse and the basis question. MOD 307 is recommended: this course inherits its habit of checking what a column contains before using it. Matrix algebra to inversion.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

A multi-regional input-output table records who supplies whom, across countries and sectors, for the whole world at once. That makes it the natural instrument for three questions a national table cannot answer: how much of a country's exports is really its own value added, where its inputs actually come from, and what a regional supply chain could plausibly substitute for. This course answers all three from implemented models — and then reads the derived indicators carefully, because three of them turn out to mean something other than their labels say.

Four of this course's laboratories are column screens, because four of its findings are about columns. One indicator is capped and the cap is invisible in the finished figure; one column is a constant reference value that reads as a variable; and the two tables that would together produce a friction-weighted substitutability disagree by a factor of a thousand with neither recording its units. The course publishes the two analyses separately and does not combine them, which is the deliverable rather than a shortfall. All 51 registered findings are in the appendix.

The rule the course is built on

A column name is a claim, not a definition. Every table in this course is read before it is used: the distribution of each column, its range, its units where they are recorded and the silence where they are not. Four findings came out of that reading and none of them would have survived to the figure stage.

Who it is for

Analysts working on global value chains, trade in value added and supply-chain exposure, and statisticians who must decide whether two published tables may be joined.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	The global table and its inverse	01–02	01	4.8
P2	Value added in trade, and how it is decomposed	03–05	02–03	8.0
P3	Indicators, caps and value-chain friction	06–08	04–05	8.5
P4	Joining, structure, comparison and reporting	09–12	06–08	11.7
	Assignment and assessment time			2.0
	Total	12	8	35.0

The 12 sessions and 8 laboratories, in teaching order, and the subject matter of each:

P1 The global table and its inverse — Sessions 01–02 · 4.8 h · Laboratory 01

- **01 What an MRIO Is** — The national table, and what it cannot see; The multi-regional structure; Flat against block-diagonal layouts; The indexing convention, and getting it wrong; What an MRIO costs: size, vintage and reconciliation; The tables this school carries
- **02 The Global Leontief Inverse** — A and L for a multi-region table; What a global inverse contains that a national one does not; Computational cost, and why it is precomputed once; Reading a block of the inverse; The basis question, again — MOD 302's M-12; Verification: does $L(I - A)$ return the identity?
- *Lab 01 The Table and the Framework* — What table is this, and what decomposition framework does it use?

P2 Value added in trade, and how it is decomposed — Sessions 03–05 · 8.0 h · Laboratory 02–03

- **03 Value Added in Trade** — Gross exports against value-added exports; Domestic and foreign value added; Double counting, and where it comes from; The DVA share as a position indicator; What a high foreign share means and does not; The value-added summary for this school's economies

- **04 Decomposition Frameworks** — Koopman, Wang and Wei, and what it did; The Borin–Mancini correction; Source and sink approaches; World and exporter perspectives; Country, sector and bilateral levels; Why the framework must be stated with the number
- **05 Regional Sourcing** — The three-way split of intermediate inputs; Checking that the split closes; Reading the table by country and by sector; What a low intra-regional share means; Landlockedness, distance and the sourcing pattern; The spread across this school’s 47 economies
- *Lab 02 Value Added in Exports* — How much of each economy’s exports is its own value added?
- *Lab 03 Regional Sourcing* — Where do inputs actually come from?

P3 Indicators, caps and value-chain friction — Sessions 06–08 · 8.5 h · Laboratory 04–05

- **06 Substitution Potential** — The question, stated precisely; Supply against imports, sector by sector; The substitutability ratio; What a capped indicator does; The seven sectors at the cap; Recomputing the uncapped ratio
- **07 A Capped Indicator** — $\min(\text{supply} / \text{imports}, 1.0)$, and what it discards; Seven of seventeen at the cap; MIN at 4.37 and TEX at 1.10, both reported as 1.0000; When a cap is right and when it is not; What the uncapped ratio shows; Reporting a capped indicator honestly
- **08 Value-Chain Friction** — The weighted friction measure; The sea-import reference; The delivered wedge, and the identity behind it; One constant across seventeen sectors; Why a derived column is not an independent quantity; Verifying an identity before using a column
- *Lab 04 The Capped Indicator* — What does the substitutability measure not tell you?
- *Lab 05 Verify the Identity* — Is the delivered wedge an independent quantity?

P4 Joining, structure, comparison and reporting — Sessions 09–12 · 11.7 h · Laboratory 06–08

- **09 Joining Tables** — Two tables, two shared quantities; A ratio of exactly 1,000, seventeen times over; Neither table records its units; What a join would produce; The check: verify shared columns before joining; Why this error produces plausible output
- **10 Structure and Centrality** — Composition, concentration and the Herfindahl index; Backward and forward linkages, normalised; The four-way structural classification; Network centrality: strength, eigenvector, betweenness; What centrality adds over a linkage; The structural profile of this school’s focus economy
- **11 Comparing Economies** — The comparison set; Concentration across economies; Value-added shares across economies; Classification counts across economies; What differs because of structure and what because of resolution; The comparison that is not available — M-10
- **12 Reporting MRIO Work** — What a reader needs to know about the table; The basis question, stated once and clearly; The decomposition framework, stated with every number; The identity and unit checks, reported; How MRIO results are oversold; The MRIO note template
- *Lab 06 Check Before You Join* — Do two tables agree about the quantities they share?
- *Lab 07 Structure and Centrality* — What does a national block of a global table look like?
- *Lab 08 The MRIO Note* — What can honestly be said from this table?

What a graduate can do

1. Set out the structure of a multi-regional input-output table.
2. Explain what an MRIO adds over a national table, and what it costs.
3. Compute and interpret the global Leontief inverse.
4. Decompose gross exports into domestic and foreign value added.
5. State what the Borin–Mancini correction fixes in the KWW decomposition.
6. Read a regional sourcing table for domestic, intra-regional and extra-regional shares.
7. Compute a substitution potential and say what a capped indicator cannot tell you.

8. Verify an accounting identity before using a derived column.
9. Check that quantities shared between two tables agree before joining them.
10. Read network centrality and structural classification on a national block.
11. Compare structural indicators across economies without over-reading them.
12. Write an MRIO note that states the table, the basis, the decomposition and the checks.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 120 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35.04 %	8 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The table and the framework; Screen before you rank; Check before you join; The MRIO note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete review of a multi-regional analysis — this material, or one the learner is responsible for — delivered as an eight-section MRIO note plus the screening scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 100.04 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data and no household record; the tables are published multi-regional accounts. Nothing in the course repairs the source: the cap stays on the capped column, the constant column stays constant, neither table is rescaled to match the other, and the empty comparison column stays empty. Where a ranking is needed the course recomputes it from the source columns and labels it as recomputed.

Materials and set-up

A 108-page Technical Manual, 13 lecture decks of 149 slides, 8 laboratory decks of 102 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 120-item quiz bank, 5 assessment files, 19 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The laboratories run in Python with pandas against tables that ship with the course; no model is needed. The largest table wants about four gigabytes of memory to hold its inverse, and the Laboratory Manual gives the reduced case for machines that cannot.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in COURSE_VERSION.json.

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